

Amitabh Choudhury

+91-9140488658 | choudhryamitabh@gmail.com | [linkedin.com/in/amitabhchoudhury](https://www.linkedin.com/in/amitabhchoudhury) | github.com/amitabh1609 | leetcode.com/amitabhchoudhury19999 | personal-portfolio-cyan-eight-50.vercel.app

Summary

Data Engineer with 2 years of Snowflake/dbt experience, specializing in ML/GenAI data systems, production LLM applications, RAG systems, and MLOps platforms — targeting Machine Learning Engineer and GenAI Engineer roles. Owns 25+ production ETL/ELT pipelines (Snowflake, dbt) serving ~75 stakeholders; led a 2,025-line SAS-to-Snowflake migration across 5 data domains with zero-defect cutover. B.Tech AI/ML (CGPA 8.55). LeetCode Knight — max rating 1923, global rank 283/29,865 (top 1%).

Work Experience

Caterpillar Bengaluru, India

Data Engineer Jun 2025 – Present

Promoted from Systems Data Support Analyst within 12 months based on technical delivery.

- Own and operate 25+ production ETL/ELT pipelines via SnapLogic and dbt serving ~75 warehouse and supply-chain stakeholders; pipelines refresh as frequently as every 20 minutes, processing hundreds to millions of rows daily; results surface in Power BI dashboards used for shipment-allocation decisions.
- Led technical implementation of the MineStar MSU SAS-to-Snowflake migration: refactored 2,025 lines of legacy SAS into 43 Snowflake objects across 5 data domains (inventory, demand, backorders, schedules, forecasts for ~5,500 globally-sourced parts); consolidated 102 work tables to 32 via CTE-based optimization; validated zero-defect cutover via row-by-row parity reconciliation.
- Designed a Goods Receipt dbt pipeline with SCD Type-2 history capture at supplier/PO grain; 13 schema and relationship tests enforce data contracts; Snowflake Dynamic Tables auto-refresh summary aggregates every 30 minutes.
- Identified SQL safety gaps in an internal Snowflake Cortex LLM-powered NL-to-SQL prototype — no write protection, no adversarial evaluation, no hallucination detection; rebuilt open-source with AST-based guardrails, adversarial benchmarking, and semantic similarity hallucination detection via sentence-transformers — see Text-to-SQL project below.
- Built a Streamlit operations app for the Tidal monitoring team supporting 24/7 shift handovers; categorises critical jobs by shift, surfaces runbook-style remediation steps, persists incident metadata in a Snowflake audit table, and auto-generates shift-transition emails.

Caterpillar Bengaluru, India

Systems Data Support Analyst Jun 2024 – Jun 2025

- Delivered ad hoc analytical SQL across Snowflake for warehouse, procurement, and finance stakeholders; translated supply-chain data into insights that fed procurement-planning discussions.
- Initiated a multi-year warehouse parts usage analysis using Python and pandas to surface seasonal demand patterns; built a Random Forest demand-forecasting prototype — findings carried forward into the Data Engineer role.
- Supported and extended production Snowflake/SnapLogic/Tidal pipelines alongside senior engineers; handled data-quality triage and validation to ensure reliable refreshes for downstream consumers.

Caterpillar Bengaluru, India

Software Engineer Intern Jun 2023 – Sep 2023

- Implemented KS-test + PSI drift detection on 3 production Tidal data streams; tuned alert thresholds against 6 months of historical data, reducing false-positive alerts by ~30% and cutting unnecessary pipeline re-runs escalated to the data quality team.

Skills

Languages: Python, SQL, C++

Data Engineering: ETL/ELT, data pipeline orchestration, data modeling, Snowflake (Dynamic Tables, Tasks, Streams, Cortex), dbt (SCD Type-2, snapshots, tests), Apache Iceberg, DuckDB, Dagster, Soda Core, SnapLogic, Tidal

ML / MLOps: LightGBM (LambdaMART, Quantile), MLflow, Evidently AI, scikit-learn, FastAPI, instructor, Pydantic v2, sqlparse, SQLAlchemy

AI / GenAI: LLM application development, RAG (Retrieval-Augmented Generation), prompt engineering, hallucination detection, vector databases (Qdrant), BM25/RRF hybrid search, RAGAS, Langfuse, Claude Sonnet API, sentence-transformers (BGE-large, MiniLM-L6)

Analytics & BI: Pandas, Power BI

Projects

Text-to-SQL with Guardrails & Hallucination Detection | [GitHub](#)

FastAPI · PostgreSQL 15 · Claude Sonnet · instructor · Pydantic v2 · sqlparse · sentence-transformers · Docker

- NL-to-SQL system over a supply chain PostgreSQL database; sqlparse AST guardrails (token-type checking, not regex) hard-block DDL/DML, multi-statement injection, and catalog access; all execution runs inside PostgreSQL SET TRANSACTION READ ONLY as a database-enforced fallback. **100% guardrail block rate on all 10 seeded injection tests; zero unsafe queries executed.**
- **100% execution accuracy** on a 50-question hand-curated adversarial benchmark (JOINS, CTEs, date arithmetic, ranking, 6 unanswerable queries); back-translation hallucination detection via MiniLM-L6 cosine similarity; multi-query validation generates two independent SQL approaches and surfaces divergence; 6 Architecture Decision Records capture design trade-offs.

Supply Chain Demand Forecasting MLOps | [GitHub](#)

LightGBM · MLflow · Evidently AI · FastAPI · Streamlit · Docker · GitHub Actions

- End-to-end demand forecasting platform for 500 spare-parts SKUs rebuilt from a Caterpillar prototype; QuantileLightGBM (P10/P50/P90) via walk-forward CV achieves **34.9% WAPE and 82.6% interval coverage** (46% over best baseline); versioned Parquet feature store with 33 hand-crafted features (feature engineering: lag, rolling stats, seasonality, stockout-adjusted); FastAPI model serving layer returns P10/P50/P90 with model-version headers.
- MLflow model registry with **promotion gate** (candidate must beat production WAPE by >2%); Evidently AI separates data drift and concept drift with two different automated responses; GitHub Actions CI/CD pipeline triggers automated retraining on drift alerts; APAC demand-shock demo mirrors real Caterpillar supply-chain disruption scenario.

Hybrid RAG System with Evaluation Suite | [GitHub](#) | [Live Demo](#)

BGE-large · BM25/RRF · Qdrant · RAGAS · Claude Sonnet · Langfuse · GitHub Actions · Docker · Render

- Production RAG pipeline over Snowflake and dbt documentation; dense retrieval (BGE-large, Qdrant) fused with BM25 sparse retrieval via RRF (k=60); BGE cross-encoder reranker re-scores top candidates; Claude Haiku query rewriter expands terse questions before retrieval; answers grounded with cited source chunks; deployed live on Render.
- 50-question adversarial benchmark across 4 tiers including precision traps for Snowflake/dbt misconceptions; best configuration achieves **0.86 RAGAS faithfulness** on conceptual queries; CI eval gate in GitHub Actions fails the build on faithfulness regression; 3-config ablation study quantifies reranking vs. hybrid retrieval contribution; all retrievals traced in Langfuse.

GitHub Archive Lakehouse | [GitHub](#)

Apache Iceberg · DuckDB · Dagster · Soda Core · MinIO · Streamlit · Docker

- Bronze/Silver/Gold Iceberg lakehouse ingesting **7.4M+ GitHub Archive events**; manifest-tracked crash-resumable pipeline handles partial MinIO writes; Silver deduplicates by event ID, parses 15 event-type payloads into typed columns, routes ~600 malformed rows to a quarantine table with error reason.
- **36 Soda Core data quality checks** across all layers gate each Dagster run; Software-Defined Assets with DailyPartitionsDefinition provide full lineage in Dagster UI; GitHub Actions CI/CD pipeline fails on >3 check failures; Iceberg time-travel demonstrated on a 3.8M-row table; 5 documented failure scenarios with exact recovery commands.

Achievements

- **LeetCode Knight** — max rating 1923; global rank 283/29,865 in Weekly Contest 366 (top 1%) | amitabhchoudhury19999
- **Codeforces** — max rating 1800 (Expert) | amitabh1208
- **CodeChef** — max rating 1711 (4★) | amitabh1208
- **Centurion Hackathon (National Level)** — Team Lead; placed 13th / 500+ participants.

Certifications

- Claude Certified Architect – Foundations (CCA-F) — In progress, Anthropic, 2025

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru — B.Tech in Artificial Intelligence and Machine Learning, CGPA 8.55/10 2020 – Apr 2024